Cauê S. Borlina

Contact Information

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Education

Massachusetts Institute of Technology (MIT), Cambridge, Massachusetts, USA
Ph.D., Planetary Science, 2022

University of Michigan, Ann Arbor, Michigan, USA
B.S.E. with Honors, *summa cum laude*, Aerospace Engineering, 2016
Minor, Physics, 2016

Employment

2025-present, Gerald H. & Sharon D. Krockover New Frontiers Assistant Professor
Department of Earth, Atmospheric, and Planetary Sciences, Purdue University
Department of Physics & Astronomy by Courtesy, Purdue University

2024-2025, Assistant Professor
Department of Earth, Atmospheric, and Planetary Sciences, Purdue University

2022-2024, Blaustein Postdoctoral Fellow
Department of Earth and Planetary Sciences, Johns Hopkins University

2016-2022, Graduate Student Researcher
Department of Earth, Atmospheric and Planetary Sciences, MIT

2015, Planetary Chemistry and Astrobiology Intern
Planetary Science, NASA Jet Propulsion Laboratory

2014, Undergraduate Research Fellow
Division of Geological and Planetary Sciences, California Institute of Technology

2013-2016, Assistant in Research
Department of Climate and Space Sciences and Engineering, University of Michigan

Teaching

Assistant Professor, Department of Earth, Atmospheric, and Planetary Sciences, Purdue University
EAPS 400: Planetary Science Senior Capstone
EAPS 302: Great Issues in Space Exploration
EAPS 591: Magnetic Fields in Planetary Systems

Teaching Assistant, Department of Earth, Atmospheric, and Planetary Sciences, MIT
12.002: Introduction to Geophysics and Planetary Science
12.400: Our Space Odyssey: An Introduction to the Origins of the Solar System

Undergraduate Instructional Aide, Department of Aerospace Engineering, University of Michigan
AE 423: Computational Methods for Aerospace Engineering

Mission Involvement

2013-2016, Scientific Collaborator for the NASA Mars Science Laboratory

Honors and Awards

2026, Seed for Success Acorn Award (PI), Purdue University

2025, Seed for Success Acorn Award (Co-I), Purdue University

2025, Gerald H. & Sharon D. Krockover New Frontiers Assistant Professorship, Purdue University

2024, 2023 Outstanding Reviewer for Geochemistry, Geophysics, Geosystems

2023, Honorable Mention, 2021/2022 Nininger Meteorite Award

2021, Blaustein Postdoctoral Fellowship, Johns Hopkins University

2021, NSF Earth Sciences Postdoctoral Fellowship (declined)

2019, School of Science Quality of Life Grant (*$500 for MIT EAPS staff/student lecture series*), MIT

2018, Shrock Fellowship, MIT

2017, Mars Science Laboratory Extended Mission-1 Science and Operations Team, NASA Group Achievement Award

2017, Grayce B. Kerr Fellowship, MIT

2016, Presidential Graduate Fellowship Award, MIT

2016, Distinguished Leadership Award, University of Michigan

2015, Mars Science Laboratory Prime Mission Science and Operations Team, NASA Group Achievement Award

2012-2016, Dean's List, University of Michigan (*Fall 2012, Winter 2013, Fall 2013, Winter 2014, Fall 2014, Winter 2015, Fall 2015, Winter 2016*)

2012-2016, University Honors, University of Michigan (*Fall 2012, Winter 2013, Fall 2013, Winter 2014, Fall 2014, Winter 2015, Fall 2015, Winter 2016*)

2014, Summer Undergraduate Research Fellowship, California Institute of Technology

2014, James B. Angell Scholar, University of Michigan

2013, Aviation Week Twenty20s, Raytheon/Aviation Week

2013, Engineering Leadership Recognition, Student Space Systems Fabrication Laboratory

Service and Outreach

2026-present, Faculty Advisor, Boilers Robotics Club

2025-present, Associate Editor, Geophysical Research Letters

2025-present, Member, EAPS Alumni Committee

2024-present, Member, EAPS Graduate Committee

2023-present, Community Reviewer for Geomagnetism and Paleomagnetism, Frontiers in Earth Science

2025, Session Chair, *Workshop on Bennu and Ryugu: Samples from the Early Solar System*, Lunar and Planetary Institute

2024-2025, Member, EAPS Community and Belonging Committee

2024, Organizer, Magnetism, Atmosphere and Life Conference

2026, Convener, *From Micrograins to Planets: Advances in Fundamental Rock Magnetism* Session, AGU Fall Meeting

2022-2024, Convener, *Planetary Magnetism and Protoplanetary Disk Magnetism* Session, AGU Fall Meeting

2022-2024, Chair, Land Acknowledgement Working Group, Department of Earth and Planetary Sciences, Johns Hopkins University

2022-2023, Member, Organizational Team for the Space Telescope Science Institute Spring Symposium 2023

2023, Panelist, NASA Review Panel (FINESST)

2022, Executive Secretary, NASA Review Panel (HGIO)

2022, Convener, *Chondritic Components: CAIs and Chondrules* Session, Lunar and Planetary Science Conference

2021, Convener, *Geomagnetic Changes at the Human Timescale* Session, AGU Fall Meeting

2019-2022, Committee Member, Planetary Lunch Series, MIT

2019-2021, Committee Member, Department of Earth, Atmospheric and Planetary Sciences Faculty Search, MIT

2019-2020, Co-founder, Minds Across Generations

2019-2020, Committee Chair, EAPS Active Talks Series, MIT

2019-2020, Committee Member, Department of Earth, Atmospheric and Planetary Sciences Taskforce 2023, MIT

2016, Board Member, School of Engineering Honors Program Student Advisory Board, University of Michigan

2015-2016, President, Students for Exploration and Development of Space, Univer-

sity of Michigan

2014-2016, Outreach Committee Chair, American Institute of Aeronautics and Astronautics, University of Michigan

2013-2016, Internal Vice President, Sigma Gamma Tau (National Honor Society for Aerospace Engineering), University of Michigan

Peer Reviewer

Earth and Planetary Science Letters
Geochemistry, Geophysics, Geosystems
Geophysical Research Letters
Geology
Journal of Geophysical Research: Planets
Meteoritics & Planetary Science
Nature Communications
Science
Science Advances
Scientific Reports

Funding

2026-2029, NSF Structure and Physics of the Solid Earth, Co-PI (*$66,218*). Collaborative Research: Obtaining Paleomagnetic Time-Series Records from Corals.

2025-2028, NASA Solar System Workings, PI (*$1,100,067*). Determining the Influence of Impacts on the Lunar Magnetic History.

2025-2029, NASA Planetary Science Enabling Facilities, Co-I (*$245,787* to Co-I Borlina out of *$5,220,380*). Purdue Resources Empowering Coordinated Investigations for Sample Exploration (PRECISE).

2025, Purdue Instructional Equipment Grant, PI (*$50,000*). Mapping Magnetic Anomalies through Aeromagnetometry in Earth and Planetary Science Courses.

2025, Purdue Office of Engagement, Community Engagement, Research and Impact Grant, Co-PI (*$29,182*). Co-Design of Natural Environment VR Experience in Partnership with the Indiana Department of Natural Resources.

2025, Purdue Experiential Education, PI (*$5,000*). Enhancing Accessibility to Caves through Virtual Reality in Collaboration with the Classroom.

Students and Postdoc Mentorship

Postdocs

2026-present, Regupathi Angappan

2025-2026, Katie Bristol (NSF Postdoctoral Fellow, now at the Institute for Rock Magnetism, University of Minnesota)

Ph.D. Students

2024-present, Sedinam Biassey-Bogart

2024-present, Melissa de Andrade Nunes (University of São Paulo, Brazil)

Master's Students

2025-present, Emma Kostecki

2025-present, Roshini Saravanan

Ph.D. Committee

2026-present, Deniz Kacan (Physics & Astronomy)

2025-present, Abigail Dunnigan

2025-present, Samuel Harris

2025-present, Jae St. Pierre

2024-present, Mariana Blanco-Rojas

Undergraduate Students

2026-present, Gabriel Chambers

2026-present, Katrina Boland

2026-present, Elizabeth Gray

2025-present, Hiya Samanta

2024-present, Brianna Petruccelli

2025-2026, Arin Bhave

2024-2025, Allison Loper

2025, Henry Lee

2025, Siya Jariwala

2023-2024, Thomas Huelskamp (Johns Hopkins University)

2021-2022, Valeria Gutierrez (MIT)

2019-2020, Hannah Ledford (MIT)

2018, Fuming Chang (MIT)

FIELDWORK

2023, Minas Gerais Peruaçu Cave System and Mato Grosso Cave System, Brazil

2017, Southern Cross Domain, Australia

COMPUTER ALLOCATIONS

2025-present, Determining the Existence of Crustal Magnetism on Icy Moons
PI, ACCESS Discover Allocation

2024-present, Effect of Core-Mantle Boundary Heat Flux on the Geodynamo
Co-I, ACCESS Discover Allocation

2021, The Role of Magnetic Fields in Regulating Planetary Atmospheres
XSEDE Supercomputer Startup Allocation

Journal Articles

Advised by Borlina†

In Preparation

[28] **Borlina, C. S.**, Thompson, M. S., Crow, C. A., Tikoo, S. M., Jung, J.-I., Mansbach, E. N., Fu, R. R., Prissel, K., Prissel, T. C., Asimow, P. D., Hu, J. Characterizing the Rock Magnetic Properties of Lunar Zircons.

[27] Biassey-Bogart†, S., **Borlina, C. S.**, Rudolph, M. L., Raymond, C. A. Determining the Effects of Magnetic Pollution in Europa's Ice Shell on Magnetic Field Measurements.

[26] Kostecki†, E., **Borlina, C. S.**, Feinberg, J. M., Senan, S., Sekhon, N., Ibarra, D. E., Timbal, M. E. M., Yambing, J., Jost, A. B., McGee, D., Meduri, D., Biggin, A. J., Mason, S., Davies, C. J. Understanding the Local Behavior of the Laschamp Excursion Through a Stalagmite from the Philippines.

[25] de Andrade Nunes†, M., Petruccelli†, B., Trindade, R. I. F., **Borlina, C. S.**, Silva, G. G., Uieda, L., Neto, A. S., Yamaguti, R. The Role of Lightning Magnetization on the Martian Crust.

[24] Angappan†, R., **Borlina, C. S.**, Stanley, S., Hofstadter, M. The Role of Magnetic Fields in the Formation of Giant Planets.

Submitted

[23] **Borlina, C. S.**, Bristol†, K. E., Doctor, R., Feinberg, J. M. Bennu's Returned Samples Recorded a Solar Nebula Magnetic Field.

Published

[22] **Borlina, C. S.**, Weiss, B. P., Bai, X.-N., Tung, P. Y., Harrison, R., Mansbach, E. N., Lima, E. A., Chatterjee, N., Tissot, F. L. H., McKeegan, K. D. Paleomagnetic Evidence for a Nebular Magnetic Field from Calcium-Aluminum-rich Inclusions. *Proceedings of the National Academy of Sciences of the United States of America*, 2026.

[21] Alonso Tagle, M. L., Maggiolo, R., Gunell, H., **Borlina, C. S.**, . . . , Van Doorsselaere, T. Evolution of Atmospheric Oxygen Escape from Earth During the Last 2450 Myr. *Journal of Geophysical Research: Space Physics*, 2026.

[20] Hamilton, C. W., McEwen, A. S., . . . , **Borlina, C. S.**, . . . , Spencer, J. Comparing NASA Discovery and New Frontiers Class Mission Concepts for the Io

Volcano Observer (IVO). *The Planetary Science Journal*, 2025.

[19] **Borlina, C. S.**, Lima, E. A., Feinberg, J. M., Jaqueto, P., Lascu, I., Trindade, R. I. F., Font, E., Sánchez-Moreno, E. M., Dimuccio, L., Yokoyama, Y., Parés, J. M., Weiss, B. P., Dorale, J. A. Obtaining High-Resolution Magnetic Records From Speleothems Using Magnetic Microscopy. *Geochemistry, Geophysics, Geosystems*, 2024.

[18] Fu, R. R., Drabon, N., Weiss, B. P., **Borlina, C. S.**, Kirkpatrick, H. Statistical Reanalysis of Archean Zircon Paleointensities: No Evidence for Stagnant-Lid Tectonics. *Earth and Planetary Science Letters*, 2024.

[17] Mansbach, E. N., Weiss, B. P., Schnepf, N. R., Lima, E. A., **Borlina, C. S.**, Chatterjee, N., Gattacceca, J., Uehara, M., Wang, H. Magnetism of the Acapulco Primitive Achondrite and Implications for the Evolution of Partially Differentiated Bodies. *Journal of Geophysical Research: Planets*, 2023.

[16] Lima, E. A., Weiss, B. P., **Borlina, C. S.**, Baratchart, L., Hardin, D. P. Estimating the Net Magnetic Moment of Geological Samples From Planar Field Maps Using Multipoles. *Geochemistry, Geophysics, Geosystems*, 2023.

[15] Taylor, R., Reddy, S., Saxey, D., Rickard, W., Tang, F., **Borlina, C. S.**, Fu, R. R., Weiss, B. P., Bagot, P., Williams, H. M., Harrison, R. J. Direct Age Constraints on the Magnetism of Jack Hills Zircon. *Science Advances*, 2023.

[14] **Borlina, C. S.**, Weiss, B. P., Bryson, J. F. J., Armitage, P. J. Lifetime of the Outer Solar System Nebula From Carbonaceous Chondrites. *Journal of Geophysical Research: Planets*, 2022.

[13] **Borlina, C. S.**, Weiss, B. P., Bryson, J. F. J., Bai, X.-N., Lima, E. A., Chatterjee, N., Mansbach, E. N. Paleomagnetic Evidence for a Disk Substructure in the Early Solar System. *Science Advances*, 2021.

[12] Fu, R. R., Drabon, N., Wiedenbeck, M., Brenner, A. R., Lowe, D. R., **Borlina, C. S.** Paleomagnetism of 3.5–4.0 Ga Zircons From the Barberton Greenstone Belt, South Africa. *Earth and Planetary Science Letters*, 2020.

[11] Shang, H., Daye, M., Sivan, O., **Borlina, C. S.**, Tamura, N., Weiss, B. P., Bosak, T. Formation of Zero-Valent Iron in Iron-Reducing Cultures of *Methanosarcina barkeri*. *Environmental Science and Technology*, 2020.

[10] **Borlina, C. S.**, Weiss, B. P., Lima, E. A., Tang, F., Taylor, R. J., Einsle, J. F., Harrison, R. J., Fu, R. R., Bell, E. A., Alexander, E. W., Kirkpatrick, H., Wielicki, M. M., Harrison, T. M., Ramezani, J., Maloof, A. C. Re-Evaluating the Evidence for a Hadean-Eoarchean Dynamo. *Science Advances*, 2020.

[9] Mighani, S., Wang, H., Shuster, D. L., **Borlina, C. S.**, Nichols, C. I., Weiss, B. P. The End of the Lunar Dynamo. *Science Advances*, 2020.

[8] Tang, F., Taylor, R. J. M., Einsle, J. F., **Borlina, C. S.**, Fu, R. R., Weiss, B. P., Williams, H. M., Williams, W., Nagy, L., Midgley, P. A., Lima, E. A., Bell,

E. A., Harrison, T. M., Alexander, E. W., Harrison, R. J. Secondary Magnetite in Ancient Zircon Precludes Analysis of a Hadean Geodynamo. *Proceedings of the National Academy of Sciences of the United States of America*, 2019.

[7] Pajusalu, M., **Borlina, C. S.**, Seager, S., Ono, S., Bosak, T. Open-Source Sensor for Measuring Oxygen Partial Pressures Below 100 Microbars. *PLOS ONE*, 2018.

[6] Weiss, B. P., Fu, R. R., Einsle, J. F., Glenn, D. R., Kehayias, P., Bell, E. A., Gelb, J., Araujo, J. F., Lima, E. A., **Borlina, C. S.**, Boehnke, P., Johnstone, D. N., Harrison, T. M., Harrison, R. J., Walsworth, R. L. Secondary Magnetic Inclusions in Detrital Zircons From the Jack Hills, Western Australia and Implications for the Origin of the Geodynamo. *Geology*, 2018.

[5] **Borlina, C. S.**, Renno, N. O. The Impact of a Severe Drought on Dust Lifting in California's Owens Lake Area. *Scientific Reports*, 2017.

[4] Harrison, R. G., Barth, E., Esposito, F., Merrison, J., Montmessin, F., Aplin, K. L., **Borlina, C. S.**, Berthelier, J. J., Déprez, G., Farrell, W., Houghton, I. M. P., Renno, N. O., Nicoll, K. A., Tripathi, S. N., Zimmerman, M. Applications of Electrified Dust and Dust Devil Electrodynamics to Martian Atmospheric Electricity. *Space Science Reviews*, 2016.

[3] Martínez, G. M., Fischer, E., Renno, N. O., Sebastián, E., Kemppinen, O., Bridges, N., **Borlina, C. S.**, Meslin, P.-Y., Genzer, M., Harri, A.-M., Vicente-Retortillo, A., Ramos, M., de la Torre Juárez, M., Gómez, F., Gomez-Elvira, J., and the REMS team. Likely Frost Events at Gale Crater: Analysis From MSL/REMS Measurements. *Icarus*, 2016.

[2] **Borlina, C. S.**, Ehlmann, B. L., Kite, E. S. Modeling the Thermal and Physical Evolution of Mount Sharp's Sedimentary Rocks, Gale Crater, Mars: Implications for Diagenetic Minerals on the MSL Curiosity Rover Traverse. *Journal of Geophysical Research: Planets*, 2015.

[1] Martínez, G. M., Renno, N. O., Fischer, E., **Borlina, C. S.**, Hallet, B., de la Torre Juárez, M., Vasavada, A. R., Ramos, M., Hamilton, V., Gomez-Elvira, J., Haberle, R. M. Surface Energy Budget and Thermal Inertia at Gale Crater: Calculations From Ground-Based Measurements. *Journal of Geophysical Research: Planets*, 2014.

Invited Talks

[34] University of Arizona, Department of Geosciences, 2026.

[33] Arizona State University, Buseck Center for Meteorite Studies, 2026.

[32] Indiana University, Department of Earth and Atmospheric Sciences, 2026.

[31] University of Colorado Boulder, Department of Geological Sciences, 2025.

[30] Indiana University, Indianapolis, Department of Earth and Environmental Sciences, 2025.

[29] NASA Jet Propulsion Laboratory, Planetary Science Seminar Series, 2025.

[28] California Institute of Technology, Petrology Reading Group, 2025.

[27] University of Minnesota, Department of Earth & Environmental Sciences Seminar Series, 2025.

[26] Arizona State University, Geophysics Seminar Series, 2024.

[25] University of Maryland, Department of Astronomy, 2024.

[24] University of Michigan, Star and Planet Formation Discussion Group, Department of Astronomy, 2024.

[23] University of Massachusetts Lowell, Lowell Center for Space Science & Technology Seminar, 2024.

[22] University of Maryland, Planetary and Exoplanetary Astronomy Lunch Seminar, 2023.

[21] Smithsonian Institution, 2023.

[20] The Ohio State University, School of Earth Sciences, 2023.

[19] Purdue University, Department of Earth, Atmospheric, and Planetary Sciences, 2023.

[18] Magnetics Information Consortium Workshop, 2023.

[17] Cornell University, Department of Astronomy, 2023.

[16] Rice University, Department of Earth, Environmental and Planetary Sciences, 2023.

[15] Georgia Institute of Technology, School of Earth & Atmospheric Sciences, 2023.

[14] Lehigh University, Department of Earth and Environmental Sciences, 2023.

[13] Space Telescope Science Institute, Exoplanets/Star & Planet Formation Seminar, 2022.

[12] Johns Hopkins University Applied Physics Laboratory, Brownbag Series, 2022.

[11] International Space Science Institute, *How Heavy Elements Escape the Earth: Past, Present, and Implications to Habitability* Team Meeting, 2022.

[10] Lunar and Planetary Institute Seminar Series, 2022.

[9] Carnegie Earth and Planets Laboratory, Astronomy Seminar Series, 2022.

[8] Brown University, Department of Earth, Environmental and Planetary Sciences, 2022.

[7] University of California, Santa Cruz, Institute of Geophysics and Planetary Physics Seminar Series, 2021.

[6] University of California, Santa Cruz, Planetary Lunch Seminar, 2021.

[5] Johns Hopkins University, Department of Earth and Planetary Sciences, 2021.

[4] NASA Jet Propulsion Laboratory, Planetary Science Seminar Series, 2021.

[3] University of Hawaii, Manoa, Institute of Geophysics and Planetology, 2020.

[2] Smithsonian Institution, Department of Mineral Sciences, 2020.

[1] University of São Paulo, Department of Geophysics, 2018.

Selected Conference Presentations and Proceedings

Talks*, Invited Talks, Advised by Borlina†**

[71] Biassey-Bogart*†, S., **Borlina, C. S.**, Nimmo, F., Angappan†, R. Detecting Magnetic Orbital Signatures from Ancient Dynamos Recorded in the Interiors of Icy Moons. *AGU Fall Meeting*, 2026.

[70] Bristol*†, K. E., **Borlina, C. S.**, Feinberg, J. M. Characterizing the Magnetic Carriers in Calcium-Aluminum-rich Inclusions from CO Chondrites: Implications for Paleointensity Studies of the Protoplanetary Disk. *AGU Fall Meeting*, 2026.

[69] **Borlina, C. S.**, Thompson, M. S., Crow, C. A., Tikoo, S. M., Jung, J.-I., Mansbach, E. N., Fu, R. R., Prissel, K., Prissel, T. C., Asimow, P. D., Hu, J. Characterization of Magnetic Inclusions in Lunar Zircon Grains and Implications for Paleomagnetic Studies of the Early Moon. *AGU Fall Meeting*, 2026.

[68] Saravanan*†, R., **Borlina, C. S.**, Smith, E. F., Tucker, M. S., Ridgway, K., Eddy, M. P. Investigating the Paleomagnetic Records of Ediacaran Rhyolites from the Catoctin Formation in Pennsylvania. *AGU Fall Meeting*, 2026.

[67] Kostecki*†, E., **Borlina, C. S.**, Feinberg, J. M., Senan, S., Sekhon, N., Ibarra, D. E., Timbal, M. E. M., Yambing, J., Jost, A. B., McGee, D., Meduri, D., Biggin, A. J., Mason, S., Davies, C. J. Understanding the Local Behavior of the Laschamp Excursion Through a Stalagmite from the Philippines. *AGU Fall Meeting*, 2026.

[66] Jung, J.-I., Tikoo, S. M., Dutta, S., Asimow, P. D., Hu, J., Luo, V. Q., Hensley, A., **Borlina, C. S.** Limited Effects of Shock Remanence on Paleointensity Estimates in Lunar Mare Basalts at Low Shock Pressures. *AGU Fall Meeting*, 2026.

[65] Samanta*†, H., Bhave†, A., Kouvaras Ostrowski, A., **Borlina, C. S.**, Bramson, A. M., Ilie, R. Co-Designing a Realistic and Immersive Virtual Reality Cave Environment for Geoscience Education Through Multi-Stakeholder Engagement. *AGU Fall Meeting*, 2026.

[64] **Borlina*, C. S.**, Bristol†, K. E., Doctor, R., Feinberg, J. M. Evidence of a Solar Nebula Magnetic Field in Samples Returned from Bennu. *Annual Meeting of the Meteoritical Society*, 2026.

[63] Bristol*†, K. E., **Borlina, C. S.** Early Solar System Magnetic Records in Calcium-Aluminum-rich Inclusions from the ALHA 77307 CO Chondrite. *Annual Meeting of the Meteoritical Society*, 2026.

[62] Thompson, M. S., Bedford, C. C., **Borlina, C. S.**, Bryant, R. N., Caffee, M. W., Eddy, M. P., Horgan, B. H. N., Ickert, R. B., Pearce, B. K. D., Prissel, K., Prissel, T. C., Tremblay, M. M., Vannier, H. N., Wiens, R. C. PRECISE: A NASA Planetary Science Enabling Facility for Coordinated Sample Analysis at Purdue University. *Annual Meeting of the Meteoritical Society*, 2026.

[61] Biassey-Bogart*†, S., **Borlina, C. S.**, Rudolph, M. L., Raymond, C. A. Using Magnetic Signatures to Investigate the Internal Dynamics of Icy Moons. *Lunar and Planetary Science Conference*, 2026.

[60] **Borlina*, C. S.**, Bristol†, K. E., Doctor, R., Feinberg, J. M. Evidence of a Solar Nebula Magnetic Field in Samples Returned from Bennu. *Lunar and Planetary Science Conference*, 2026.

[59] **Borlina*, C. S.**, Bristol†, K. E., Doctor, R., Feinberg, J. M. Potential Evidence of a Solar Nebula Magnetic Field in Samples Returned From Bennu. *AGU Fall Meeting*, 2025.

[58] Biassey-Bogart*†, S., **Borlina, C. S.**, Rudolph, M. L. Using Magnetic Signatures to Investigate the Internal Dynamics of Icy Moons. *AGU Fall Meeting*, 2025.

[57] Petruccelli*†, B., **Borlina, C. S.**, de Andrade Nunes†, M., Trindade, R. I. F. Investigating the Role of Lightning in Remanent Magnetization on the Surface of Mars. *AGU Fall Meeting*, 2025.

[56] Bristol†, K. E., **Borlina, C. S.**, Feinberg, J. M. From Bulk Rock Magnetism to Individual Inclusions: A Detailed Magnetic Characterization of Refractory Inclusions in Chondrites for Paleomagnetic Analysis. *AGU Fall Meeting*, 2025.

[55] Kouvaras Ostrowski, A., **Borlina, C. S.**, Samanta, H., Jariwala, S., Lee, H., Bramson, A. M., Ilie, R. Developing and Co-Designing a Cave Virtual Reality Experience for Students in Geoscience Courses. *AGU Fall Meeting*, 2025.

[54] Martin, C., Lavier, L., Taylor, F. W., **Borlina, C. S.** Obtaining Paleomagnetic Time-series Records From Corals. *AGU Fall Meeting*, 2025.

[53] **Borlina*, C. S.**, Bristol†, K. E., Doctor, R., Feinberg, J. M. Potential Evidence of a Solar Nebula Magnetic Field in Samples Returned From Bennu. *Workshop on Bennu and Ryugu: Samples from the Early Solar System*, 2025.

[52] Bristol*†, K. E., **Borlina, C. S.**, Feinberg, J. M. Magnetized in Brines: Evidence For Robust Magnetic Carriers From Bennu's Returned Samples. *Workshop on Bennu and Ryugu: Samples from the Early Solar System*, 2025.

[51] Thompson, M. S., Bedford, C. C., **Borlina, C. S.**, Bryant, R. N., Caffee, M. W., Eddy, M. P., Horgan, B. H. N., Ickert, R. B., Pearce, B. K. D., Prissel, K.,

Prissel, T. C., Tremblay, M. M., Wiens, R. C. Purdue Resources Empowering Coordinated Investigations for Sample Exploration (PRECISE): A New NASA Planetary Science Enabling Facility. *Annual Meeting of the Meteoritical Society*, 2025.

[50] Bristol†, K. E., **Borlina, C. S.**, Feinberg, J. M. Magnetic Characterization of Chondrites and their Suitability for Paleomagnetic Analysis of Refractory Inclusions. *IRM Conference on Rock Magnetism*, 2025.

[49] Biassey-Bogart†, S., **Borlina, C. S.**, Rudolph, M. L. Using Magnetic Signatures to Investigate the Internal Dynamics of Icy Moons. *IRM Conference on Rock Magnetism*, 2025.

[48] **Borlina*, C. S.**, Tikoo, S. M., Crow, C. A., Mansbach, E. N., Fu, R. R. Using Lunar Zircon Grains to Elucidate the Early Lunar Dynamo. *Lunar and Planetary Science Conference*, 2025.

[47] Brain, D., Berta-Thompson, Z., Chaffin, M., . . . , **Borlina, C. S.**, Frelikh, R., Notsu, Y., Thomas, T. Atmospheric Loss from Venus-like Planets Orbiting G Stars. *American Astronomical Society Meeting*, 2025.

[46] **Borlina, C. S.**, Tikoo, S. M., Crow, C. A., Mansbach, E. N. Determining the Potential of Lunar Zircons as Paleomagnetic Targets. *AGU Fall Meeting*, 2024.

[45] Lascu, I., Jaqueto, P., Livi, K., Feinberg, J. M., **Borlina, C. S.**, Lima, E. A. Towards a Clearer Picture of Speleothem Magnetism Carriers: In Situ Imaging of Magnetic Particles from Calcite and Aragonite Stalagmites. *AGU Fall Meeting*, 2024.

[44] Alonso Tagle, M. L., Maggiolo, R., Gunell, H., Cessateur, G., De Keyser, J., Dandouras, I., Vidotto, A., **Borlina, C. S.**, Nichols, C., Lapenta, G., Pierrard, V., Vandaele, A. C. Evolution of Atmospheric Oxygen Escape on Earth since the Paleoproterozoic Era. *EGU General Assembly Conference*, 2024.

[43] **Borlina**, C. S.**, Weiss, B. P., Bai, X.-N., Mansbach, E. N., Lima, E. A., Chatterjee, N., Tissot, F. L. H., McKeegan, K. D. Paleomagnetic Evidence for a Nebular Magnetic Field at the Beginning of the Solar System. *Goldschmidt Conference*, 2024.

[42] **Borlina, C. S.**, Stanley, S., Hofstadter, M. How Planetary Magnetic Fields Influence the Formation of Giant Planets. *AGU Fall Meeting*, 2023.

[41] Tung, P., Harrison, R. J., **Borlina, C. S.**, Weiss, B. P. SIGMA: A Machine Learning Tool for Identifying and Unmixing Mineral Phases from Energy Dispersive X-ray Spectroscopy (EDS) Data. *AGU Fall Meeting*, 2023.

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